

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 608 Database Systems — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
CMU 15-445/645 Database Systems — Andy Pavlo (free video, slides, projects)	The primary source. Complete lectures, slides, and the BusTub project assignments. The best free database systems course that exists, by some margin.
CMU 15-721 Advanced Database Systems (free)	Where modern in-memory and column-store systems are covered. Watch the relevant lectures after Modules 05 and 09.
MIT 6.5830 Database Systems (free)	A second voice with a different emphasis — more on query processing theory and less on implementation detail.
The Red Book — Readings in Database Systems, 5th ed. (free)	Curated classic papers with editorial commentary explaining why each mattered. Read the commentary even where you skip the paper.
Database Internals — Alex Petrov (sample chapters free)	Strong on storage engines and on distributed systems. Good complement to the CMU material for Modules 02–04 and 13.
Architecture of a Database System — Hellerstein, Stonebraker & Hamilton (free)	A 100-page tour of how the components fit together. Read it in week one and again at the end.

Module-by-module map

Module 01 — What a Database Is For

Resource	What it covers for this module
CMU 15-445 — Lecture 1, Course Introduction and Relational Model	Pavlo's opening lecture covers §1 and §2 with more history and more jokes. Watch it before reading these notes.
Hellerstein, Stonebraker & Hamilton — Architecture of a Database System (free)	The map of §3. Read it this week.
Codd — A Relational Model of Data for Large Shared Data Banks (1970, free)	The original paper. Short, and the data independence argument of §2 is clearer in the original than in most retellings.
The Red Book — editorial introduction	Stonebraker and Hellerstein's commentary on what has actually mattered in fifty years of the field. Opinionated and worth reading.

Module 02 — Storage and the Buffer Pool

Resource	What it covers for this module
CMU 15-445 — Lectures 3–6, Storage and Buffer Pools	Page layout, slotted pages, and the buffer pool with replacement policies. The primary source for this module.
Crotty, Leis & Pavlo — Are You Sure You Want to Use MMAP in Your DBMS? (free)	The §4 argument in full, with measurements.
O'Neil, O'Neil & Weikum — The LRU-K Page Replacement Algorithm (free)	The original LRU-K paper, including the sequential flooding analysis.
PostgreSQL fsync reliability discussion ('fsyncgate')	The §4 story, documented by the people who found it. Worth reading as a case study in how durability assumptions fail.
Database Internals — chapters on storage engines	Page layouts and buffer management from a different angle.

Module 03 — B+Trees

Resource	What it covers for this module
CMU 15-445 — Lectures 7–8, Tree Indexes	B+tree structure, splits, merges, and the design decisions in §2. The project assignment is the implementation.
Graefe — Modern B-Tree Techniques (free)	The definitive survey. Prefix compression, suffix truncation, bulk loading, and everything else in §2's refinements list.
Use The Index, Luke (free online book)	The best free material on §3 — when indexes help, composite ordering, and covering indexes, with examples in several dialects.
Database Internals — B-tree chapters	A second treatment with more attention to on-disk layout details.

Module 04 — Hash Indexes, LSM Trees, and Concurrent Access

Resource	What it covers for this module
CMU 15-445 — Lectures 7 and 9, Hash Tables and Index Concurrency	Extendible and linear hashing, and latch crabbing with the safe-node condition.
O'Neil et al. — The Log-Structured Merge-Tree (1996, free)	The original LSM paper. Short, and the amplification argument of §2 is already in it.
Athanassoulis et al. — Designing Access Methods: The RUM Conjecture (free)	The framing in §2 that makes the trade-off explicit.
Lehman & Yao — Efficient Locking for Concurrent Operations on B-Trees (1981, free)	The B ^{link} -tree of §3. Still the basis of PostgreSQL's implementation.
RocksDB wiki	A production LSM engine, documented in detail by its maintainers. The compaction and tuning pages are the practical side of §2.

Module 05 — Relational Algebra and SQL

Resource	What it covers for this module
CMU 15-445 — Lecture 2, Modern SQL	SQL beyond the basics: window functions, CTEs, lateral joins, and the evaluation order of §3.
Date — writings on NULL and three-valued logic	The case against NULL, argued at length by someone who worked with Codd. Opinionated, and the technical content in §2 is correct.
Markus Winand — Modern SQL (free)	Excellent free material on what SQL has gained since 1992, and on the evaluation order.
PostgreSQL documentation — Queries chapter	Precise on evaluation order, NULL handling, and where the implementation differs from the standard.

Module 06 — Sorting and Joins

Resource	What it covers for this module
CMU 15-445 — Lectures 11–12, Sorting and Joins	External sort, the three join families, and the cost analysis of §2.
Graefe — Query Evaluation Techniques for Large Databases (free)	The comprehensive survey of sorting, hashing, and join processing. Dense, and still the reference.
Shapiro — Join Processing in Database Systems with Large Main Memories (free)	The Grace and hybrid hash join analysis of §2.
PostgreSQL — Using EXPLAIN	The diagnostic material in §3, with worked examples. Read this before tuning anything.

Module 07 — Query Execution

Resource	What it covers for this module
CMU 15-445 — Lectures 13–14, Query Execution	The iterator model, execution models, and parallelism. The primary source.
Graefe — Volcano, An Extensible and Parallel Query Evaluation System (free)	The iterator model of §1, in the original.
Boncz, Zukowski & Nes — MonetDB/X100: Hyper-Pipelining Query Execution (free)	The vectorised execution of §2, and the paper that established it.
Neumann — Efficiently Compiling Efficient Query Plans for Modern Hardware (free)	Query compilation, from HyPer. The counterpoint to vectorisation.
Leis et al. — Morsel-Driven Parallelism (free)	The scheduling approach of §4.

Module 08 — Query Optimisation

Resource	What it covers for this module
CMU 15-445 — Lectures 15–16, Query Planning and Optimization	Rewrites, cost estimation, and System R enumeration.
Selinger et al. — Access Path Selection in a Relational Database Management System (1979, free)	The System R paper. Dynamic programming join enumeration and interesting orders, in the original. Still worth reading.
Leis et al. — How Good Are Query Optimizers, Really? (free)	The measurement study behind §2's honest assessment. Read it.
PostgreSQL — planner statistics and extended statistics	What the optimiser knows and how to give it more, including the correlation fix in §2.
Use The Index, Luke — execution plan chapters	Reading plans across several database systems.

Module 09 — Transactions and Concurrency Control

Resource	What it covers for this module
CMU 15-445 — Lectures 17–18, Transactions and Two-Phase Locking	ACID, serializability theory, 2PL, and deadlock handling.
Berenson et al. — A Critique of ANSI SQL Isolation Levels (free)	The §4 argument, from the people who built these systems. Short and important.
Gray & Reuter — Transaction Processing: Concepts and Techniques	The standard reference. Not free, and in most university libraries; Gray's individual papers are free and cover much of it.

Resource	What it covers for this module
Martin Kleppmann — Designing Data-Intensive Applications, chapter 7	The clearest modern explanation of isolation levels and the anomalies, with real incident examples.
Jepsen — consistency model analyses (free)	Empirical testing of what systems actually provide against what they claim. The practical counterpart to §4's advice.

Module 10 — Multi-Version Concurrency Control

Resource	What it covers for this module
CMU 15-445 — Lecture 19, Multi-Version Concurrency Control	Version storage, visibility, and garbage collection across real systems.
Wu et al. — An Empirical Evaluation of In-Memory MVCC (free)	A systematic comparison of version storage schemes and their costs. The source for §3's table.
Cahill, Röhm & Fekete — Serializable Isolation for Snapshot Databases (free)	SSI, which became PostgreSQL's SERIALIZABLE. The dangerous-structure argument is clearer in the original.
PostgreSQL documentation — Concurrency Control and Routine Vacuuming	The implementation in §1 and §3, including wraparound and what to monitor.
Kleppmann — Designing Data-Intensive Applications, chapter 7	Write skew explained with more real examples, including ones that caused incidents.

Module 11 — Logging and Recovery

Resource	What it covers for this module
CMU 15-445 — Lectures 20–21, Crash Recovery	WAL, steal/force policies, and the three ARIES passes.
Mohan et al. — ARIES (1992, free)	The original paper. Long, and the repeating-history argument in §2 is worth reading in full.
PostgreSQL documentation — Write-Ahead Logging and Reliability	A production implementation, including group commit and the fsync settings that control durability.
Pillai et al. — All File Systems Are Not Created Equal (free)	What filesystems actually guarantee about ordering and atomicity, and how many applications get it wrong. Sobering, and directly relevant to §4.
Jepsen — database analyses	Empirical crash and partition testing of real systems. The standard §4 describes, applied professionally.

Module 12 — Column Stores and Analytics

Resource	What it covers for this module
CMU 15-721 — Advanced Database Systems (free)	Pavlo's graduate course, largely about the systems in this module. The column store and compression lectures are the primary source.
Abadi, Boncz & Harizopoulos — The Design and Implementation of Modern Column-Oriented Database Systems (free)	The definitive survey. Late materialisation, compression, and the execution model, with measurements.
Stonebraker et al. — C-Store (2005, free)	The paper that established the modern column store, and the ancestor of Vertica.
Apache Arrow and Parquet documentation	The open formats of §4. Worth understanding directly — they are increasingly the interface between systems.
DuckDB documentation and papers	A readable, modern column store you can install in one command and experiment with immediately.

Module 13 — Distributed Databases

Resource	What it covers for this module
Kleppmann — Designing Data-Intensive Applications	The best single book on this material. Chapters 5 through 9 cover everything in this module with more care and more examples.
Abadi — Consistency Tradeoffs in Modern Distributed Database System Design (PACELC, free)	The §2 framing, in the original. Short.
Ongaro & Ousterhout — In Search of an Understandable Consensus Algorithm (Raft, free)	Raft. The paper is readable, and the site has an animated visualisation that makes leader election obvious.
Corbett et al. — Spanner (free)	TrueTime and externally consistent distributed transactions. Read it for the clock argument.
Jepsen — analyses of real distributed databases	What systems actually do under partition, as opposed to what they claim. Essential reading before trusting any consistency claim.